

????????

Generated: 2026-08-12 20:22:47

????????????????????????????????

****A Case of Formal Construction: The Psyche of the Calculator, Foundation-Building (???)**, Infin

> ?????????????????????????

> ?????????????????

> ?? = ??????????????????delta???????????????????? = ?????????????????/????????????????

*2026-08-12 · Internal research paper · Lean 4 / mathlib v4.32.2 · 20 claims, all PROVED, no so

??

???"?????"????????????????????**????delta ?????????????????**22 ? claim ?? Lean 4 / mathlib ??????

0. ??????????"???"

????????????????????????**??**?

1. **??**RulerTwo???????? 2????????????
2. **??**RulerDelta??delta????????????????
3. **??**RulerDrift????????????????
4. **??**RulerLoss????????????
5. **??**RulerFoldCircle???? [0,?]????
6. **??**RulerFourier????????

????????????????

???????? 2?RulerTwo?

?·1 ?????????

????????8 ??? ? 4 ??? ? 2 ? ? 1????**??? 8?4?2?1 ??? 2 ???**2³?2²?2¹?2?????"?????

?·2 ???2 ?????

****Claim (RulerTwo, PROVED)**?**2 = ?????????

"???? 2"????????????????

1. **????**TK2??`sign : Perm ? ? ??`?? {+1, -1}????/???
2. **????**TK1??`transport e f x = (f -? e) +? x` ?????**2 ?"???"????**?1 ?????????2 ???
3. **????**????8?4?2?1 ????? = ?????
4. **????**Lean ???Bool = 2 ?????Prop ?????

????2 = ?????????1 ?????**2 ?????**?"?"?? ?2 ????? 2^k = k ????? = k ???

?·3 ?????????

"???? ?2 ?"***???????????? Church ???C-META13????2 ?????????????

****? (Lean)**?**`sign\_is\_pm\_one` [TK2]; `transport` [TK1]; `dofLevel`/`dofLevel\_succ` [TK4]?

??delta ?????????RulerDelta, RulerDrift, RulerAsym?

?·1 ?????????

?????????"????????????????delta?????????"????

- ****???????? e ? ?? T_{e?f} ? delta = f -? e??? = ???
- ****delta ???delta ? ??? (e, e+delta) ? e ???**? = delta ?????**?

?·2 ???delta ?????

****Claim (RulerDelta, PROVED)**?**delta????????

???????? delta???? (e, e+delta) ? e ????????? delta?? f-e = f'-e' ?????? **delta ????

?·3 ????????

Claim (RulerDrift, PROVED)??????????????

- ??value?= ????? e ???????position?= delta ???D(H)??
 - ???(e,a) ~ (f,b) ? a-e = b-f?delta ????? e?f ?????????????
 - ??? 0/1/-2 ? delta=3?? (0,3)/(1,4)/(-2,1) ????? [delta=3] ???
- ?????"?????????????"?***?? = ???delta ????? = ?????**?

?·4 ??? 2 = ??????

Claim (RulerAsym, PROVED)??? 2 ??????? 2?±1?????

- *** 2**{a,b} ????????????
- *** 2**{+1,-1} ????????????????

???±1???***??*????????????????????±1/?/????/???? vs ???*????*?C-PH2 + RulerDelta ? **?????

*** (Lean)**`transportParameter` [Ruler2Exam]; `displacement\_structure\_is\_equiv` [TK1]; `tran
????????????RulerLoss, RulerFold, RulerMiddle, RulerDivergence?

?·1 ??????????

?????????"?????????????????????"????????????????????

?·2 ?? = ??? = ??????

Claim (RulerLoss, PROVED)??? = ?????

- ???C011????????proj ?????-1 ?????????
- ?????????????????1/n ?????????
- ?????????C007 ?????? ???????1/n² ? ?²/6?
- ***?***?? = ??? = ???????

???????????????????????????????? ±????"?"??

??***????? ? ?????????????????***? ????? C011??

?·3 ????????????

Claim (RulerFold, PROVED)???????????????? ???? ? ?????? ? ??? ? ???

***???????????????? vs ???**?

- ***?lossless?***????????????????????/????
- ***?lossy?***????????????????proj ???*????????**?

***?????? = ???*?C-META14 ?????

?·4 ??/? = ????

Claim (RulerMiddle, PROVED)?-1 ?? / 0 ?? / +1 ???

- ?????????????????????***?***?***? ??? = ?????
- ??1???????????? ? ??????
- ??2????????????***tokenizer ???"?????????"**?

?·5 ?? ? ??????

Claim (RulerDivergence, PROVED)??

*** (Lean)**`proj\_mul\_not\_preserved` [C011]; `zeta\_euler\_product` [C023]; `phase\_finite` [Div

????????????RulerCycle, RulerPhase, RulerFourier, RulerSimplify, RulerFFT

?·1 ????????? = ?????

?????"????????????????????????????????"

Claim (RulerCycle, PROVED)???? = ??? ?/2 ?????

- ????? = 0, ?/2, ?, 3?/2???? = ?????

